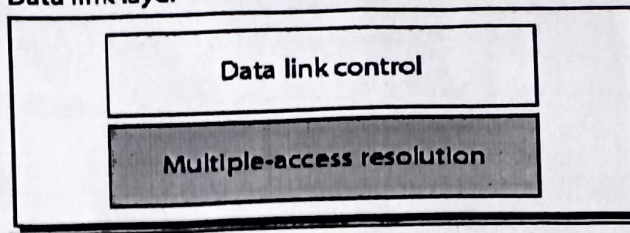
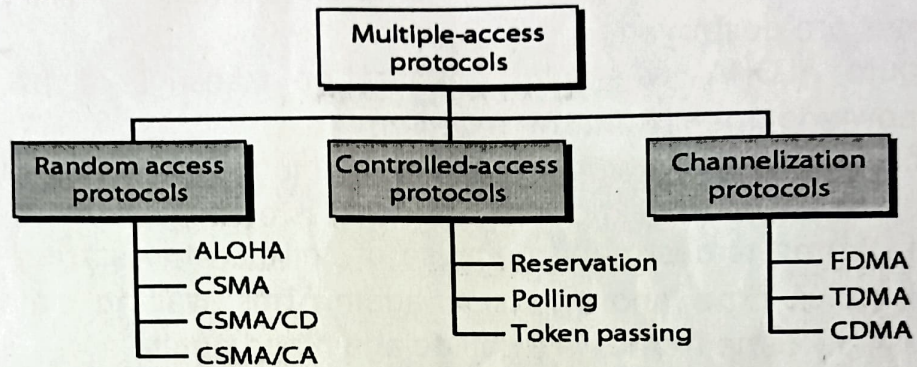


Data link layer



The upper sub layer that is responsible for flow and error control is called the logical link control (LLC) layer; the lower sub layer that is mostly responsible for multiple access resolution is called the media access control (MAC) layer. When nodes or stations are connected and use a common link, called a multipoint or broadcast link, we need a multiple-access protocol to coordinate access to the link.



Taxonomy of multiple-access protocols

RANDOM ACCESS

In random access or contention methods, no station is superior to another station and none is assigned the control over another.

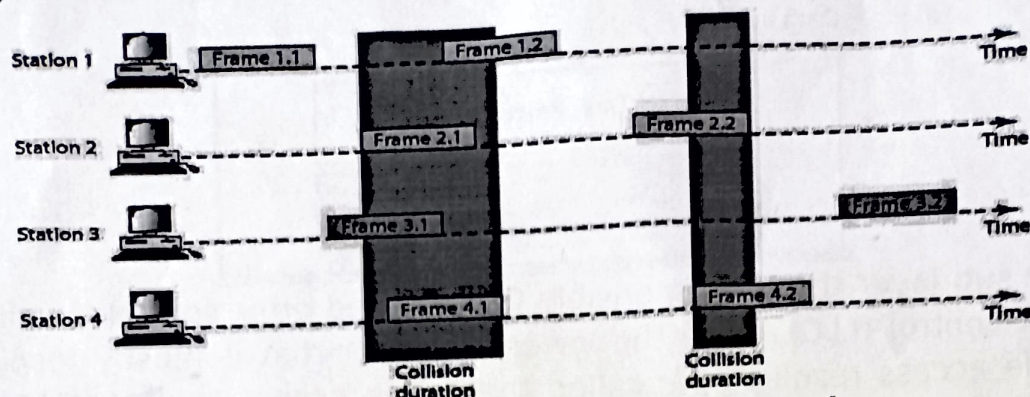
Two features give this method its name. First, there is no scheduled time for a station to transmit. Transmission is random among the stations. That is why these methods are called *random access*. Second, no rules specify which station should send next. Stations compete with one another to access the medium. That is why these methods are also called *contention* methods.

ALOHA

1 Pure ALOHA

The original ALOHA protocol is called pure ALOHA. This is a simple, but elegant protocol. The idea is that each station sends a frame whenever it has a frame to send. However, since there is only one channel to share, there is the possibility of collision between frames from different stations. Below Figure shows an example of frame collisions in pure ALOHA.

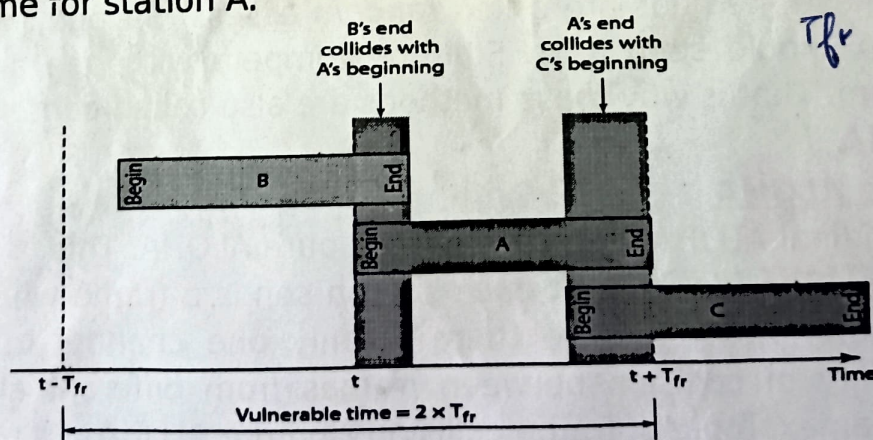
3



Frames in a pure ALOHA network

- In pure ALOHA, the stations transmit frames whenever they have data to send.
- When two or more stations transmit simultaneously, there is collision and the frames are destroyed.
- In pure ALOHA, whenever any station transmits a frame, it expects the acknowledgement from the receiver.
- If acknowledgement is not received within specified time, the station assumes that the frame (or acknowledgement) has been destroyed.
- If the frame is destroyed because of collision the station waits for a random amount of time and sends it again. This waiting time must be random otherwise same frames will collide again and again.
- Therefore pure ALOHA dictates that when time-out period passes, each station must wait for a random amount of time before resending its frame. This randomness will help avoid more collisions.

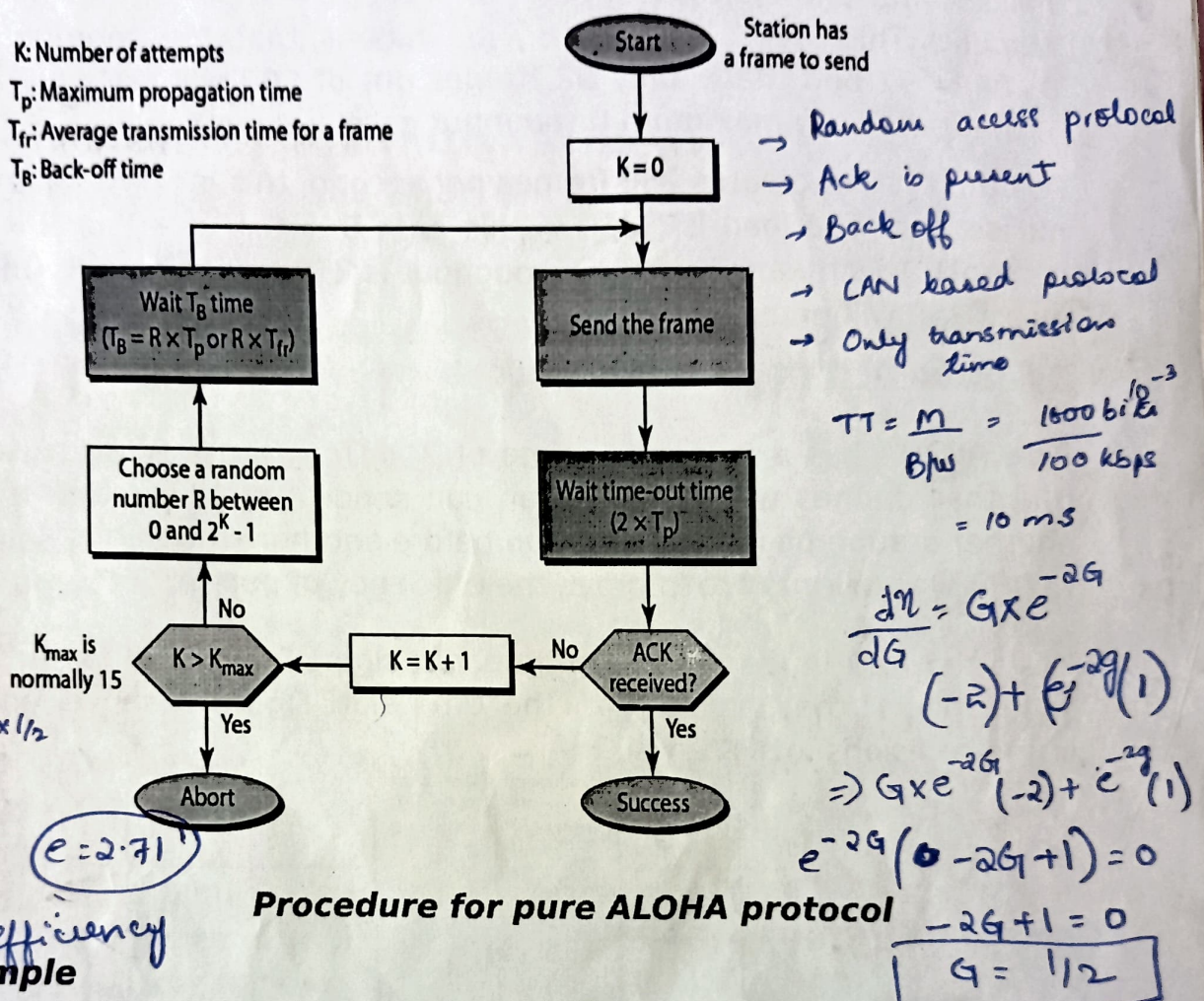
Vulnerable time Let us find the length of time, the **vulnerable time**, in which there is a possibility of collision. We assume that the stations send fixed-length frames with each frame taking T_{fr} to send. Below Figure shows the vulnerable time for station A.



$T_{fr} \rightarrow$ average frame transmission time

Station A sends a frame at time t . Now imagine station B has already sent a frame between $t - T_{fr}$ and t . This leads to a collision between the frames from station A and station B. The end of B's frame collides with the beginning of A's frame. On the other hand, suppose that station C sends a frame between t and $t + T_{fr}$. Here, there is a collision between frames from station A and station C. The beginning of C's frame collides with the end of A's frame

Looking at Figure, we see that the vulnerable time, during which a collision may occur in pure ALOHA, is 2 times the frame transmission time. Pure ALOHA vulnerable time = $2 \times T_{fr}$



Example

A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the requirement to make this frame collision-free?

Solution

Average frame transmission time T_{fr} is 200 bits/200 kbps or 1 ms. The vulnerable time is $2 \times 1 \text{ ms} = 2 \text{ ms}$. This means no station should send later than 1 ms before this station starts transmission and no station should start sending during the one 1-ms period that this station is sending.

The throughput for pure ALOHA is $S = G \times e^{-2G}$. The maximum throughput $S_{max} = 0.184$ when $G = (1/2)$. *no. of stations*

PROBLEM

A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces a. 1000 frames per second b. 500 frames per second c. 250 frames per second. The frame transmission time is 200/200 kbps or 1 ms,

a. If the system creates 1000 frames per second, this is 1 frame per

(5)

millisecond. The load is 1. In this case $S = G \times e^{-2G}$ or $S = 0.135$ (13.5 percent). This means that the throughput is $1000 \times 0.135 = 135$ frames. Only 135 frames out of 1000 will probably survive.

- b. If the system creates 500 frames per second, this is $(1/2)$ frame per millisecond. The load is $(1/2)$. In this case $S = G \times e^{-2G}$ or $S = 0.184$ (18.4 percent). This means that the throughput is $500 \times 0.184 = 92$ and that only 92 frames out of 500 will probably survive. Note that this is the maximum throughput case, percentage wise.
- c. If the system creates 250 frames per second, this is $(1/4)$ frame per millisecond. The load is $(1/4)$. In this case $S = G \times e^{-2G}$ or $S = 0.152$ (15.2 percent). This means that the throughput is $250 \times 0.152 = 38$. Only 38 frames out of 250 will probably survive.

2 Slotted ALOHA

Pure ALOHA has a vulnerable time of $2 \times T_{fr}$. This is so because there is no rule that defines when the station can send. A station may send soon after another station has started or soon before another station has finished. Slotted ALOHA was invented to improve the efficiency of pure ALOHA.

In slotted ALOHA we divide the time into slots of T_{fr} s and force the station to send only at the beginning of the time slot. Figure 3 shows an example of frame collisions in slotted ALOHA

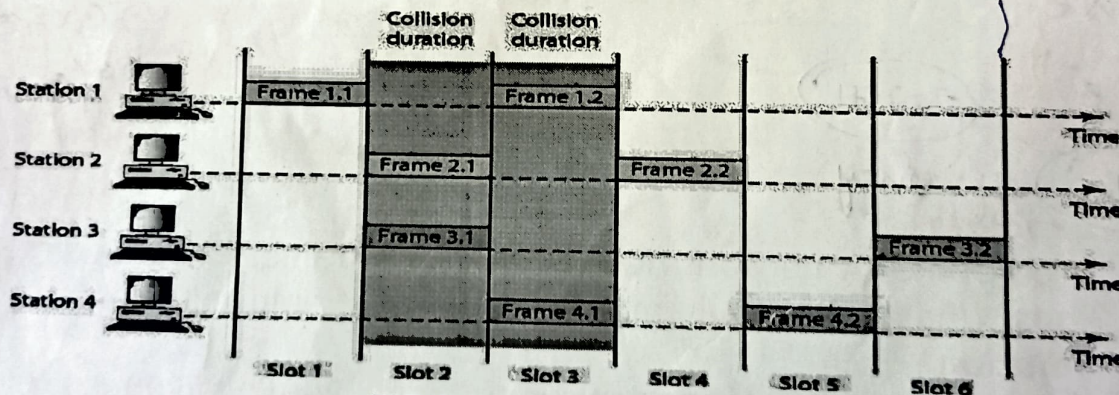
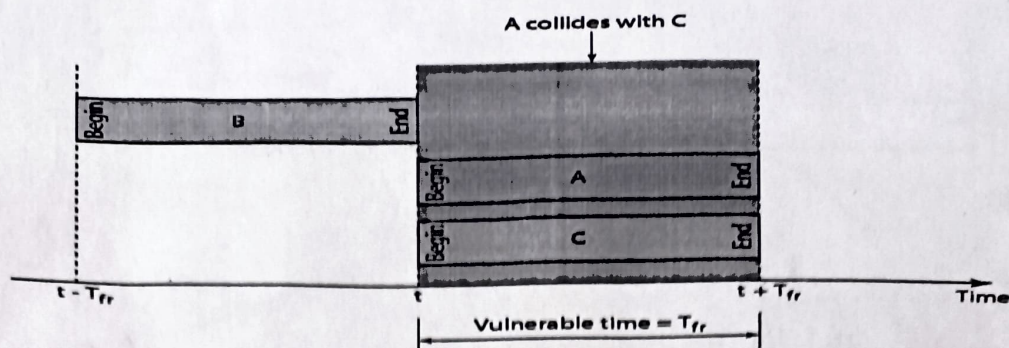


FIG:3

Because a station is allowed to send only at the beginning of the synchronized time slot, if a station misses this moment, it must wait until the beginning of the next time slot. This means that the station which started at the beginning of this slot has already finished sending its frame. Of course, there is still the possibility of collision if two stations try to send at the beginning of the same time slot. However, the vulnerable time is now reduced to one-half, equal to T_{fr} . Figure 4 shows the situation

Below fig shows that the vulnerable time for slotted ALOHA is one-half that of pure ALOHA. Slotted ALOHA vulnerable time = T_{fr}



The throughput for slotted ALOHA is $S = G \times e^{-G}$. The maximum throughput $S_{max} = 0.368$ when $G = 1$.

A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200-Kbps bandwidth. Find the throughput if the system (all stations together) produces

- a. 1000 frames per second b. 500 frames per second c. 250 frames per second

Solution

This situation is similar to the previous exercise except that the network is using slotted ALOHA instead of pure ALOHA. The frame transmission time is $200/200$ kbps or 1 ms.

- a. In this case G is 1. So $S = G \times e^{-G}$ or $S = 0.368$ (36.8 percent). This means that the throughput is $1000 \times 0.368 = 368$ frames. Only 368 out of 1000 frames will probably survive. Note that this is the maximum throughput case, percentagewise.
- b. Here G is $1/2$. In this case $S = G \times e^{-G}$ or $S = 0.303$ (30.3 percent). This means that the throughput is $500 \times 0.303 = 151$. Only 151 frames out of 500 will probably survive.
- c. Now G is $1/4$. In this case $S = G \times e^{-G}$ or $S = 0.195$ (19.5 percent). This means that the throughput is $250 \times 0.195 = 49$. Only 49 frames out of 250 will probably survive.

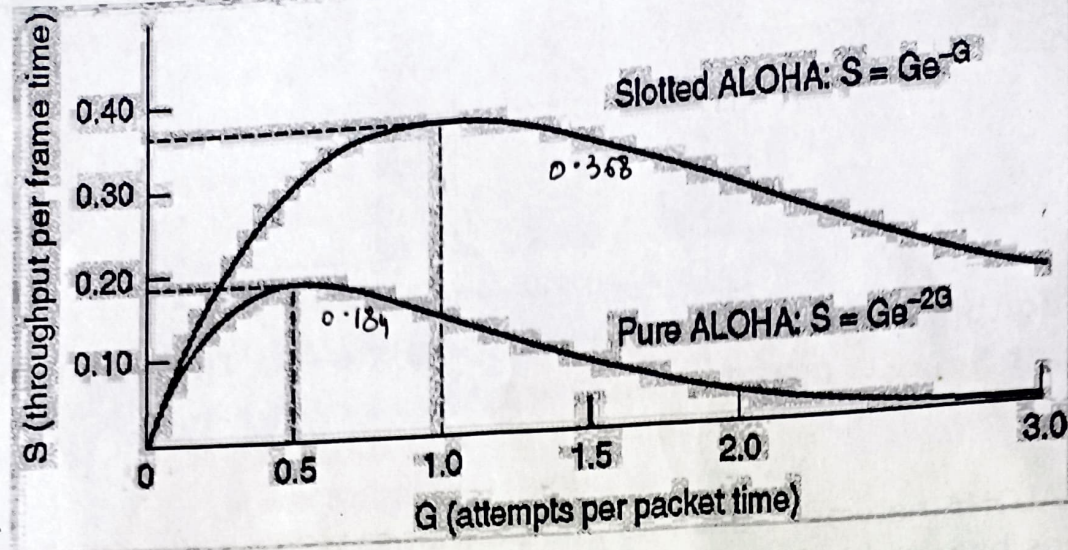
Comparison between Pure Aloha & Slotted Aloha

$$\begin{aligned}
 S &= G \times e^{-G} \\
 &= 1 \times e^{-1} \\
 &= \frac{1}{e} = \frac{1}{2.71} \\
 &= 0.368
 \end{aligned}$$

$$\begin{aligned}
 \frac{dS}{dG} &= 0 \\
 \frac{dS}{dG} &= G \times e^{-G}(-1) + e^{-G}(1) \\
 &= e^{-G}(-G + 1) = 0
 \end{aligned}$$

$$\begin{aligned}
 -G + 1 &= 0 \\
 G &= 1
 \end{aligned}$$

7

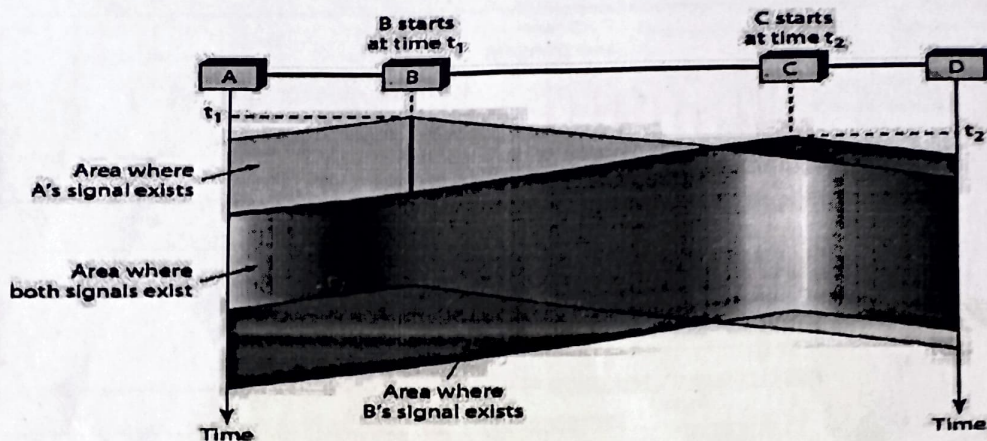


Carrier Sense Multiple Access (CSMA)

To minimize the chance of collision and, therefore, increase the performance, the CSMA method was developed. The chance of collision can be reduced if a station senses the medium before trying to use it. Carrier sense multiple access (CSMA) requires that each station first listen to the medium (or check the state of the medium) before sending. In other words, CSMA is based on the principle "sense before transmit" or "listen before talk."

CSMA can reduce the possibility of collision, but it cannot eliminate it. The reason for this is shown in below Figure. Stations are connected to a shared channel (usually a dedicated medium). The possibility of collision still exists because of propagation delay; station may sense the medium and find it idle, only because the first bit sent by another station has not yet been received.

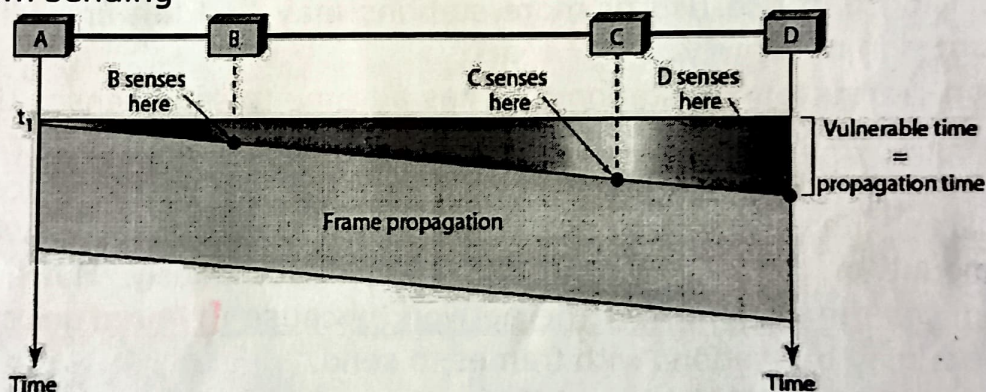
At time t_1' station B senses the medium and finds it idle, so it sends a frame. At time t_2 ($t_2 > t_1'$) station C senses the medium and finds it idle because, at this time, the first bits from station B have not reached station C. Station C also sends a frame. The two signals collide and both frames are destroyed.



Space/time model of the collision in CSMA

Vulnerable Time

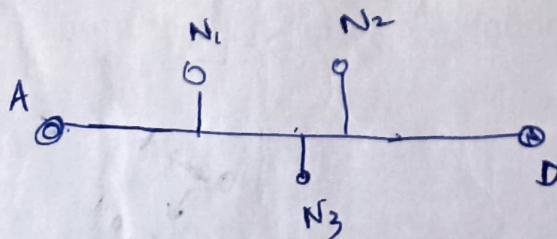
The vulnerable time for CSMA is the propagation time T_p . This is the time needed for a signal to propagate from one end of the medium to the other. When a station sends a frame, and any other station tries to send a frame during this time, a collision will result. But if the first bit of the frame reaches the end of the medium, every station will already have heard the bit and will refrain from sending.

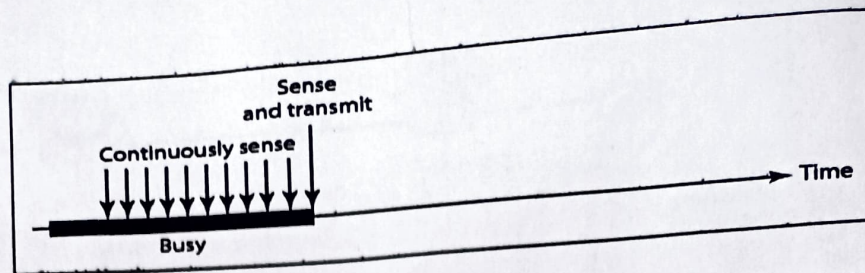


Vulnerable time in CSMA

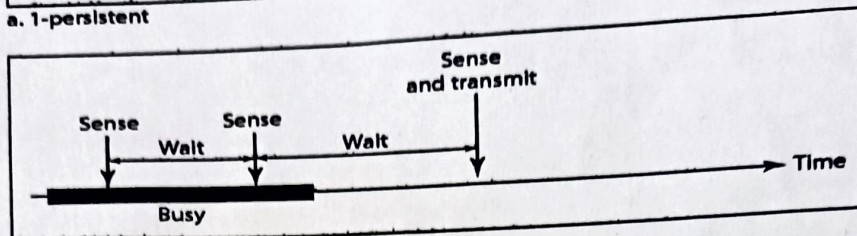
Persistence Methods

What should a station do if the channel is busy? What should a station do if the channel is idle? Three methods have been devised to answer these questions: the 1-persistent method, the non-persistent method, and the p-persistent method.

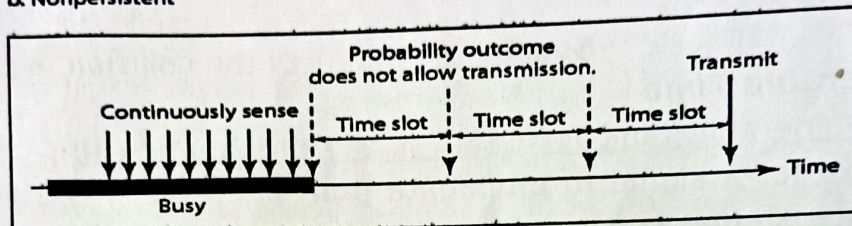




a. 1-persistent



b. Nonpersistent



c. p-persistent

1-Persistent: In this method, after the station finds the line idle, it sends its frame immediately (with probability 1). This method has the highest chance of collision because two or more stations may find the line idle and send their frames immediately.

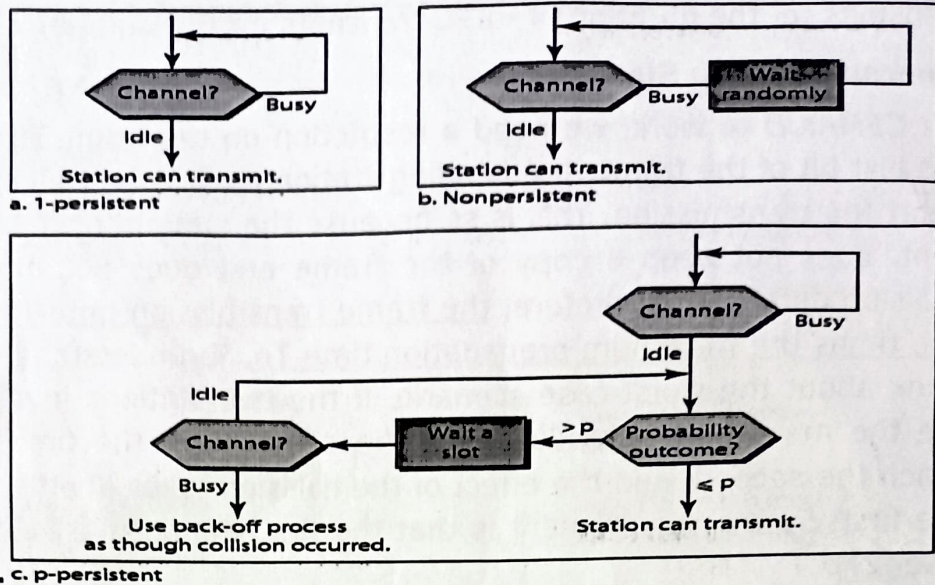
Non-persistent: a station that has a frame to send senses the line. If the line is idle, it sends immediately. If the line is not idle, it waits a random amount of time and then senses the line again. This approach reduces the chance of collision because it is unlikely that two or more stations will wait the same amount of time and retry to send simultaneously. However, this method reduces the efficiency of the network because the medium remains idle when there may be stations with frames to send.

p-Persistent: This is used if the channel has time slots with a slot duration equal to or greater than the maximum propagation time. The p-persistent approach combines the advantages of the other two strategies. It reduces the chance of collision and improves efficiency.

In this method, after the station finds the line idle it follows these steps:

1. With probability p , the station sends its frame.
2. With probability $q = 1 - p$, the station waits for the beginning of the next time slot and checks the line again.
 - a. If the line is idle, it goes to step 1.
 - b. If the line is busy, it acts as though a collision has occurred and uses the backoff procedure.

0.5
0.5
0.5

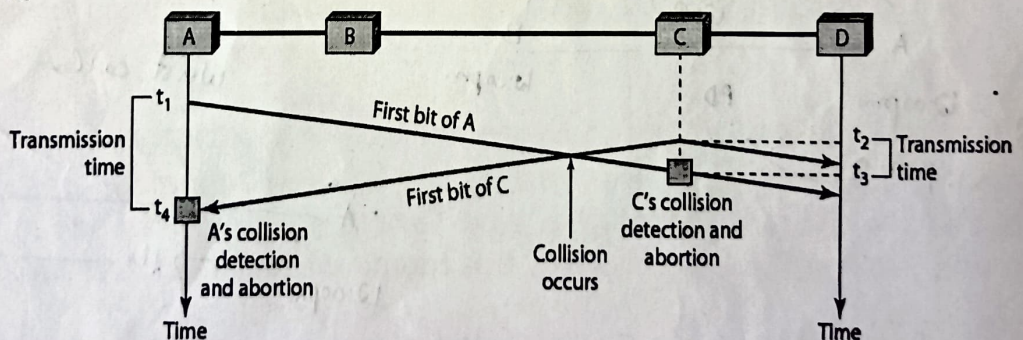


Carrier Sense Multiple Access with Collision Detection (CSMA/CD)

The CSMA method does not specify the procedure following a collision. Carrier sense multiple access with collision detection (CSMA/CD) augments the algorithm to **handle the collision**.

In this method, a station monitors the medium after it sends a frame to see if the transmission was successful. If so, the station is finished. If, however, there is a collision, the frame is sent again.

To better understand CSMA/CD, let us look at the first bits transmitted by the two stations involved in the collision. Although each station continues to send bits in the frame until it detects the collision, we show what happens as the first bits collide. In below Figure, stations A and C are involved in the collision.



Collision of the first bit in CSMA/CD

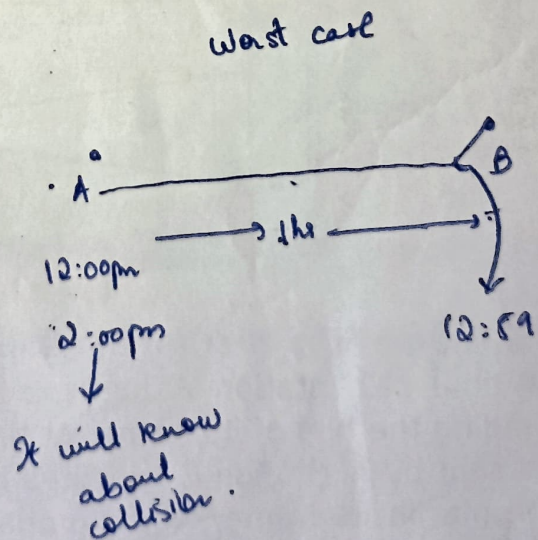
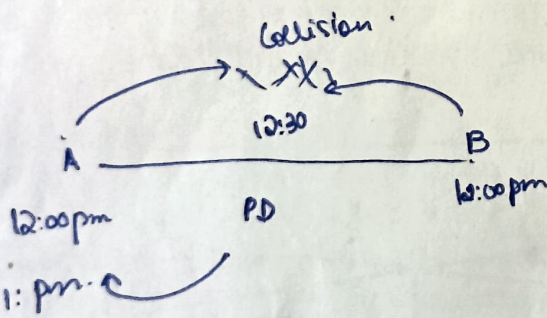
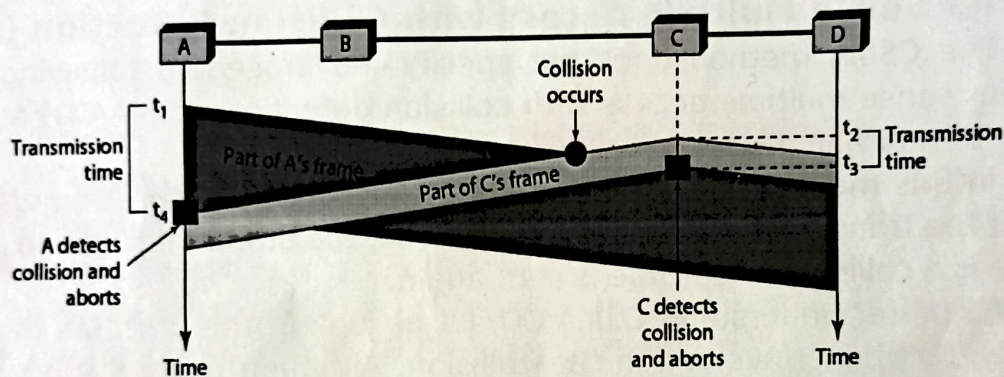
At time t_1 , station A has executed its persistence procedure and starts sending the bits of its frame. At time t_2 , station C has not yet sensed the first bit sent by A. Station C executes its persistence procedure and starts sending the bits in its frame, which propagate both to the left and to the right. The collision occurs sometime after time t_2 . Station C detects a collision at time t_3 when it receives the first bit of A's frame. Station C immediately (or after a short time, but we assume immediately) aborts transmission.

Station A detects collision at time t_4 when it receives the first bit of C's frame; it also immediately aborts transmission. Looking at the figure, we see that A

transmits for the duration $t_4 - t_1$; C transmits for the duration $t_3 - t_2$.

Minimum Frame Size

For CSMA/CD to work, we need a restriction on the frame size. Before sending the last bit of the frame, the sending station must detect a collision, if any, and abort the transmission. This is so because the station, once the entire frame is sent, does not keep a copy of the frame and does not monitor the line for collision detection. Therefore, the frame transmission time T_{fr} must be at least two times the maximum propagation time T_p . To understand the reason, let us think about the worst-case scenario. If the two stations involved in a collision are the maximum distance apart, the signal from the first takes time T_p to reach the second, and the effect of the collision takes another time T_p to reach the first. So the requirement is that the first station must still be transmitting after $2T_p$.



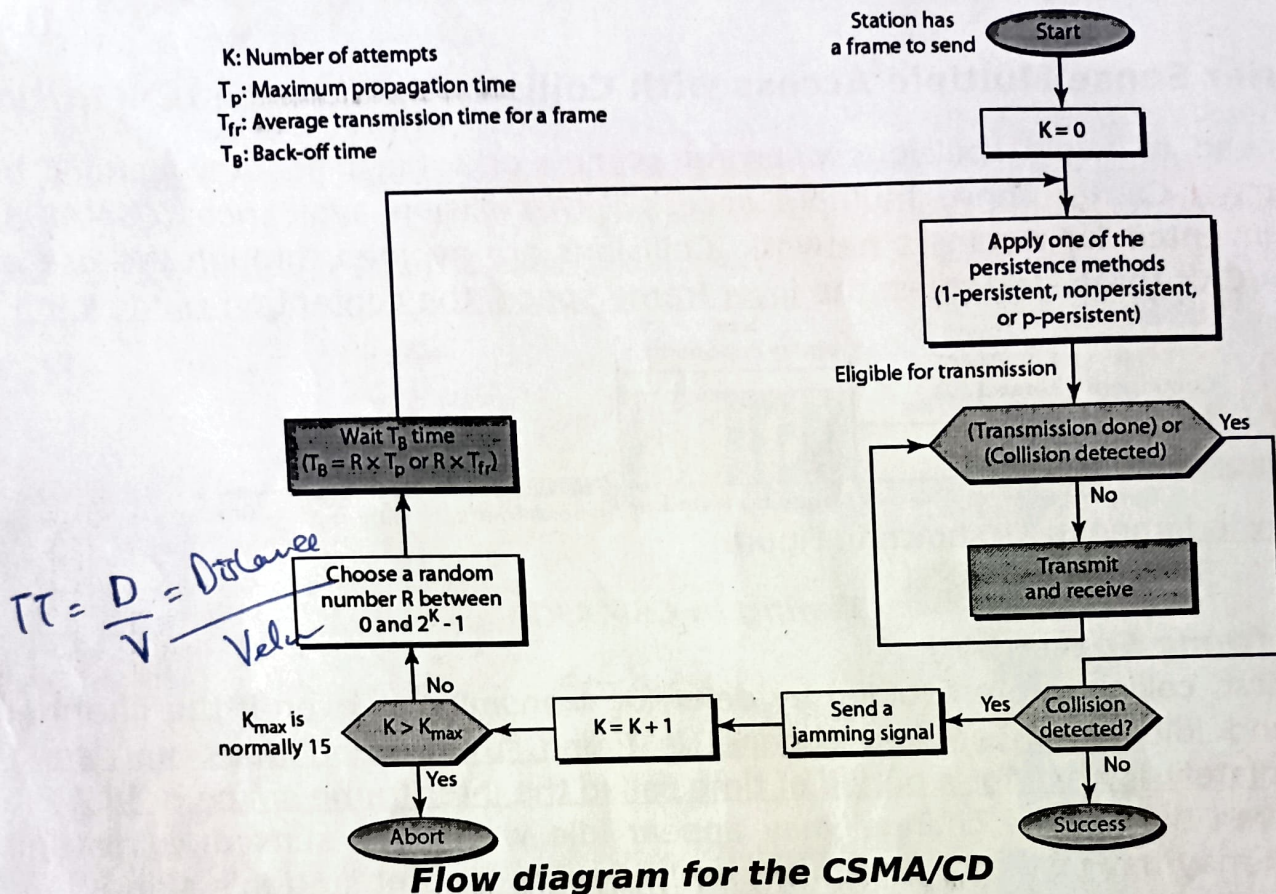
$$T_{fr} \geq 2PD$$

$$T_{fr} = \frac{\text{Message}}{BW} \geq 2PD$$

$$L \geq 2PD \times BW$$

Collision and abortion in CSMA/CD

K: Number of attempts
 T_p : Maximum propagation time
 T_{fr} : Average transmission time for a frame
 T_B : Back-off time



Flow diagram for the CSMA/CD

PROBLEM

A network using CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation time (including the delays in the devices and ignoring the time needed to send a jamming signal, as we see later) is 25.6 μ s, what is the minimum size of the frame?

SOL

The frame transmission time is $T_{fr} = 2 \times T_p = 51.2 \mu$ s. This means, in the worst case, a station needs to transmit for a period of 51.2 μ s to detect the collision. The minimum size of the frame is 10 Mbps $\times$ 51.2 μ s = 512 bits or 64 bytes. This is actually the minimum size of the frame for Standard Ethernet.

DIFFERENCES BETWEEN ALOHA & CSMA/CD

The first difference is the addition of the persistence process. We need to sense the channel before we start sending the frame by using one of the persistence processes

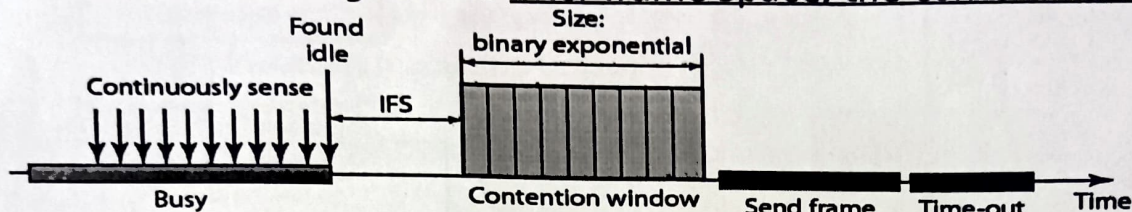
The second difference is the frame transmission. In ALOHA, we first transmit the entire frame and then wait for an acknowledgment. In CSMA/CD, transmission and collision detection is a continuous process. We do not send the entire frame and then look for a collision. The station transmits and receives continuously and simultaneously

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The third difference is the sending of a short jamming signal that enforces the collision in case other stations have not yet sensed the collision.

Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)

We need to avoid collisions on wireless networks because they cannot be detected. Carrier sense multiple access with collision avoidance (CSMA/CA) was invented for wireless network. Collisions are avoided through the use of CSMA/CA's three strategies: the inter frame space, the contention window, and



acknowledgments, as shown in Figure

Timing in CSMA/CA

Inter frame Space (IFS)

First, collisions are avoided by deferring transmission even if the channel is found idle. When an idle channel is found, the station does not send immediately. It waits for a period of time called the inter frame space or IFS.

Even though the channel may appear idle when it is sensed, a distant station may have already started transmitting. The distant station's signal has not yet reached this station. The IFS time allows the front of the transmitted signal by the distant station to reach this station. If after the IFS time the channel is still idle, the station can send, but it still needs to wait a time equal to the contention time. The IFS variable can also be used to prioritize stations or frame types. For example, a station that is assigned shorter IFS has a higher priority.

In CSMA/CA, the IFS can also be used to define the priority of a station or a frame.

Contention Window

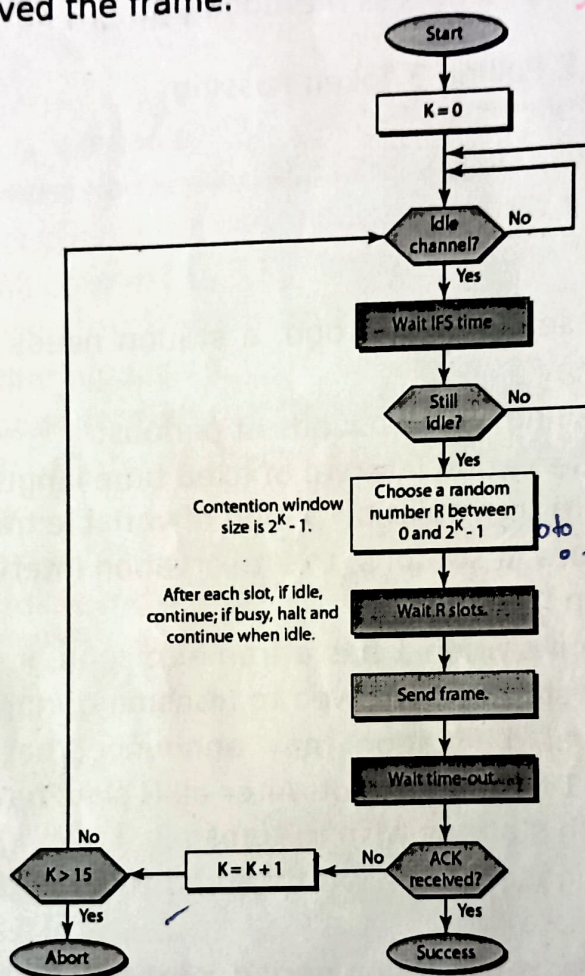
The contention window is an amount of time divided into slots. A station that is ready to send chooses a random number of slots as its wait time. The number of slots in the window changes according to the binary exponential back-off strategy. This means that it is set to one slot the first time and then doubles each time the station cannot detect an idle channel after the IFS time. This is very similar to the p-persistent method except that a random outcome defines the number of slots taken by the waiting station.

One interesting point about the contention window is that the station needs to sense the channel after each time slot. However, if the station finds the channel busy, it does not restart the process; it just stops the timer and restarts it when the channel is sensed as idle. This gives priority to the station with the longest waiting time.

In CSMA/CA, if the station finds the channel busy, it does not restart the timer of the contention window; it stops the timer and restarts it when the channel becomes idle.

Acknowledgment

With all these precautions, there still may be a collision resulting in destroyed data. In addition, the data may be corrupted during the transmission. The positive acknowledgment and the time-out timer can help guarantee that the receiver has received the frame.



This is the CSMA protocol with collision avoidance.

- The station ready to transmit, senses the line by using one of the persistent strategies.
- As soon as it finds the line to be idle, the station waits for an IFS (Inter frame space) amount of time.
- If then waits for some random time and sends the frame.
- After sending the frame, it sets a timer and waits for the acknowledgement from the receiver.
- If the acknowledgement is received before expiry of the timer, then the transmission is successful.
- But if the transmitting station does not receive the expected acknowledgement before the timer expiry then it increments the back off

parameter, waits for the back off time and re senses the line

Controlled Access Protocols

In controlled access, the stations seek information from one another to find which station has the right to send. It allows only **one node** to send at a time, to avoid collision of messages on shared medium.

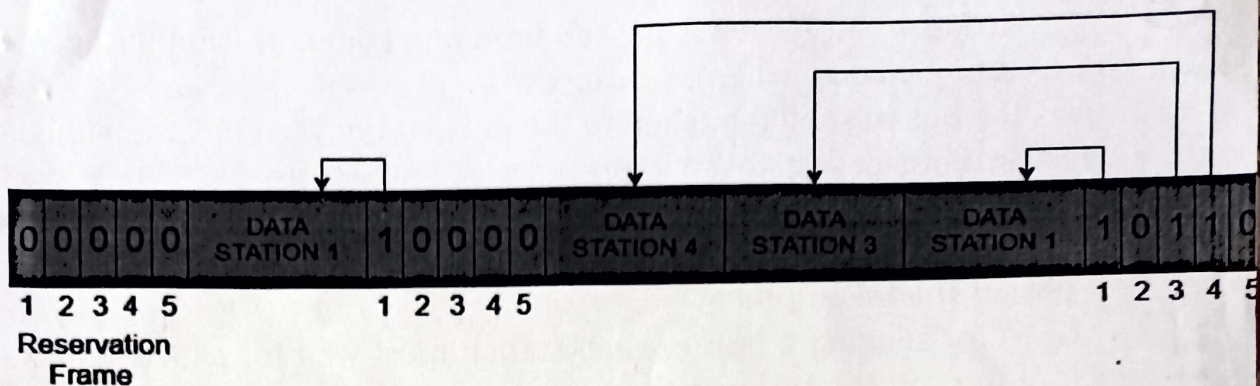
The three controlled-access methods are:

1 Reservation 2 Polling 3 Token Passing

Reservation

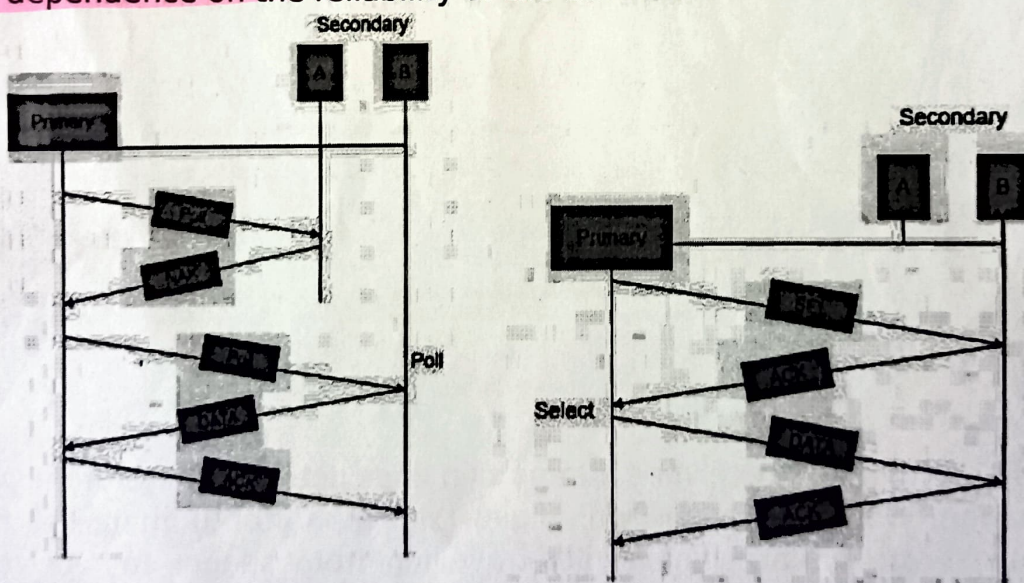
- In the reservation method, a station needs to make a reservation before sending data.
- The time line has two kinds of periods:
 1. Reservation interval of fixed time length
 2. Data transmission period of variable frames.
- If there are M stations, the reservation interval is divided into M slots, and each station has one slot.
- Suppose if station 1 has a frame to send, it transmits 1 bit during the slot
 1. No other station is allowed to transmit during this slot.
- In general, i^{th} station may announce that it has a frame to send by inserting a 1 bit into i^{th} slot. After all N slots have been checked, each station knows which stations wish to transmit.
- The stations which have reserved their slots transfer their frames in that order.
- After data transmission period, next reservation interval begins.
- Since everyone agrees on who goes next, there will never be any collisions.

The following figure shows a situation with five stations and a five slot reservation frame. In the first interval, only stations 1, 3, and 4 have made reservations. In the second interval, only station 1 has made a reservation.



Polling

- Polling process is similar to the roll-call performed in class. Just like the teacher, a controller sends a message to each node in turn.
- In this, one acts as a primary station(controller) and the others are secondary stations. All data exchanges must be made through the controller.
- The message sent by the controller contains the address of the node being selected for granting access.
- Although all nodes receive the message but the addressed one responds to it and sends data, if any. If there is no data, usually a "poll reject"(NAK) message is sent back.
- Problems include high overhead of the polling messages and high dependence on the reliability of the controller.



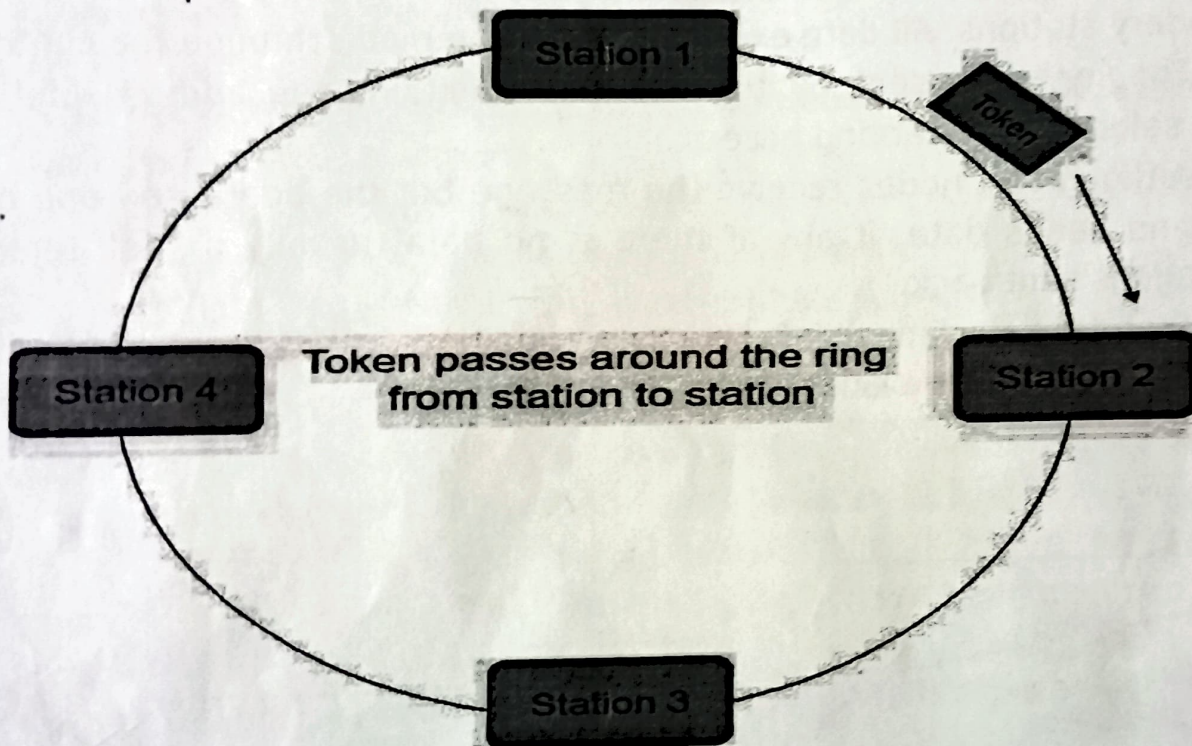
Token Passing

- In token passing scheme, the stations are connected logically to each other in form of ring and access of stations is governed by tokens.
- A token is a special bit pattern or a small message, which circulate from one station to the next in the some predefined order.

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In Token ring, token is passed from one station to another adjacent station in the ring whereas incase of Token bus, each station uses the bus to send the token to the next station in some predefined order.

- In both cases, token represents permission to send. If a station has a frame queued for transmission when it receives the token, it can send that frame before it passes the token to the next station. If it has no queued frame, it passes the token simply.
- After sending a frame, each station must wait for all N stations (including itself) to send the token to their neighbors and the other N - 1 stations to send a frame, if they have one.
- There exists problems like duplication of token or token is lost or insertion of new station, removal of a station, which need be tackled for correct and reliable operation of this scheme.

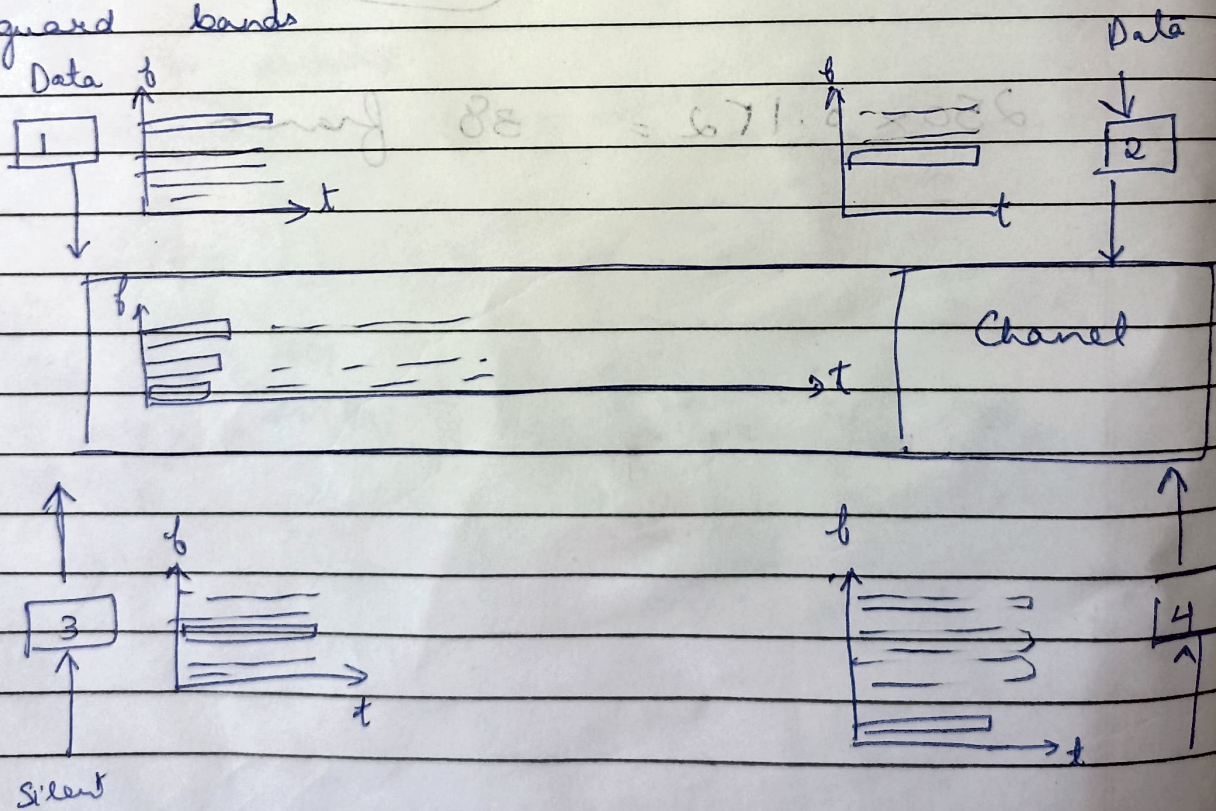


CHANNELIZATION

Channelization is a multiple access method in which the available b/w of a link is shared in time, frequency or code.

FDMA - Frequency Division Multiple access.

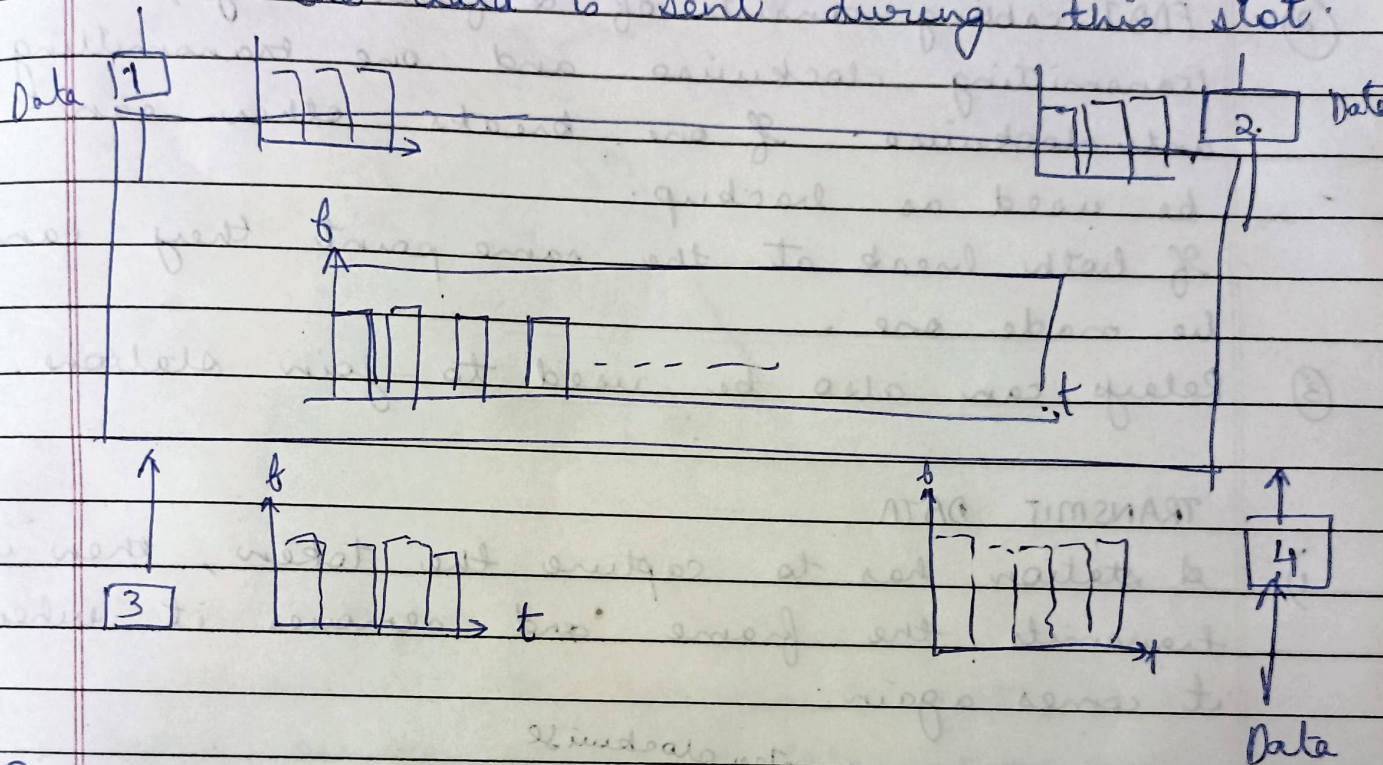
- ① The available bandwidth is divided into frequency bands. Every station is allocated a band to send its data.
- ② Every station uses a bandpass filter.
- ③ To prevent station interferences, the bands are separated from one another by guard bands.



- ④ Stream data (data having a continuous flow) can be used with FDMA because channel is available throughout.

TDMA TIME DIVISION MULTIPLE ACCESS

- ① In TDMA, stations share the bandwidth of channel in time. Every station is allocated a time slot which is used for sending data. The data is sent during this slot.



- ② The problem of synchronization between stations exists. (sync bit)
- ③ A station knows the beginning and location of its slot.

CDMA Code division multiple access

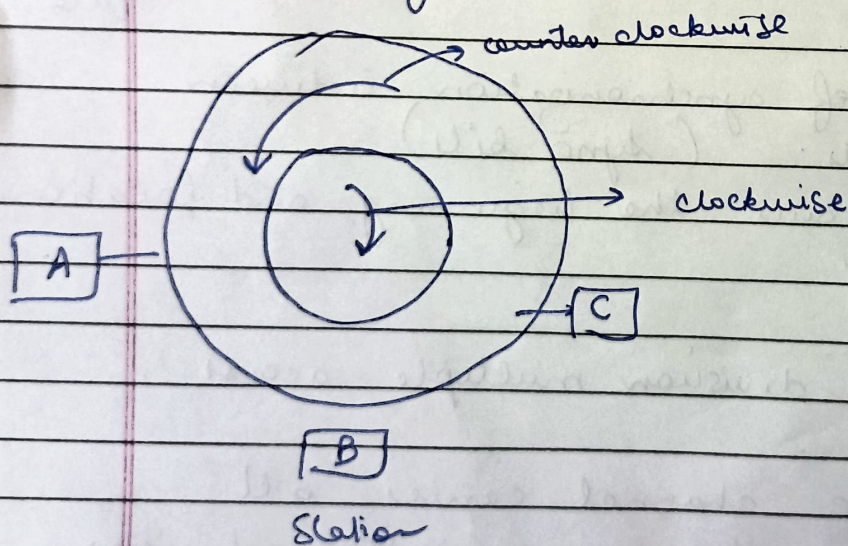
In CDMA, one channel carries all transmissions. The communication devices should exchange data in a code nobody else understands.

INTERFACE (FDDI)

- ① FDDI is a high performance fiber optic token ring.
- ② FDDI cabling consists of 2 fiber rings, one transmitting clockwise and one transmitting anticlockwise. If one breaks other can be used as backup. If both break at the same point they can be made one.
- ③ Relays can also be used to join stations

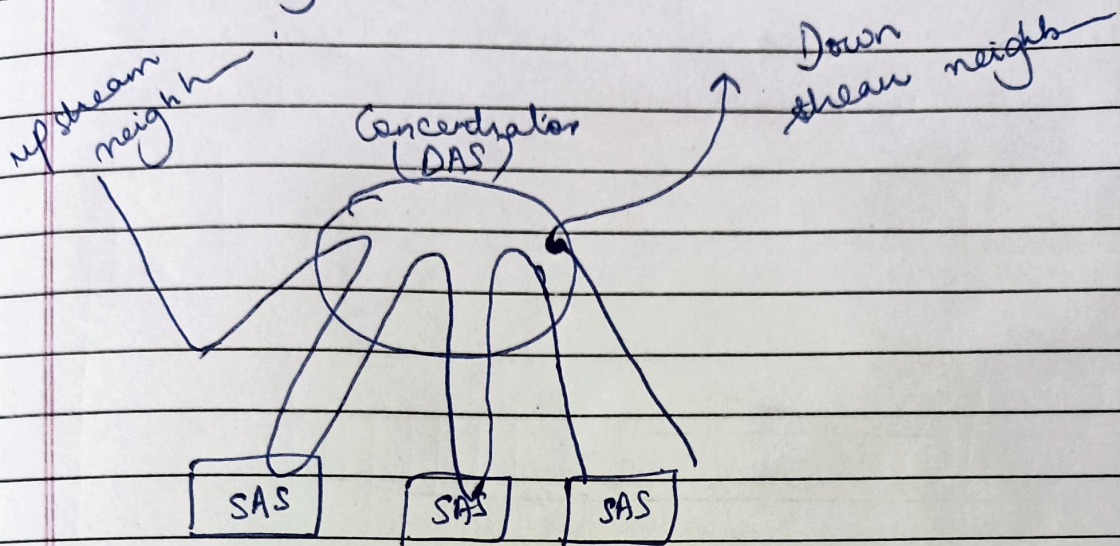
TRANSMIT DATA

- 1) A station has to capture the token, then it transmits the frame and removes it when it comes again.



FDDI allows nodes to attach to the network by a single cable. These nodes are called single attachment stations (SAS) their dual counterparts are called dual attachment stations.

A concentrator is used to attach SAS to dual ring.



TRAFFIC TYPE:

- 1) Synchronous: This data can be sent whether token arrives early or late. e.g. voice, video
- 2) asynchronous: Only if the token arrives early, this data can be sent.

TIMERS:

- ① Token holding timer: It determines how long a station may continue to transmit once a token has been acquired
- ② Token rotation timer: It is restarted everytime a token is seen. If it expires it means timer has been lost.
- ③ Valid transmission timer: Used to timeout